

The Space Station Decision

With the majority of Senior Interagency Group for Space [SIG (Space)] agencies being opposed to going ahead with a space station, according to Hans Mark by the end of August he and Jim Beggs were "quite pessimistic" about getting a positive presidential decision. That pessimism increased in the following weeks. Mark suggests that "the last weeks of September ... were the low point" in the space agency's hope to gain space station approval, as former National Aeronautics and Space Administration (NASA) administrator James Fletcher "came by for a visit and told us that we should think about backing away from pushing the space station project." Mark suggests that "the actual nadir of our fortunes" came at an October 4, 1983, staff meeting at which Beggs "wondered aloud whether Fletcher might not be right ... We really were in the position of fighting an uphill battle, and there does come a time to cut losses." This may have been Beggs musing out loud, not a considered position. At any rate, reports Mark, "while we considered backing away, we finally decided to stick to our guns."¹

Beggs was somewhat more optimistic than Mark; through his occasional interactions with President Reagan he had become convinced that Reagan wanted to move forward with a major space initiative in the civil sector, among other things as a counterbalance to the Strategic Defense Initiative (SDI) and the accompanying accusation that his administration was "militarizing space." Beggs continued between August and December 1983 to seek support for the station initiative from key individuals close to the president. For example, NASA asked Alan Shepard, the Project Mercury and Project Apollo astronaut who had been the first American in space and who by 1983 was a very successful businessman in Houston, to talk to his friend James Baker, White House chief of staff, about the space station. In his approach to managing White House activities, Baker seldom took an active role in advocating or opposing a particular policy choice, but behind the scenes he had become one of the president's most trusted policy advisers. Thus, his support for the station

initiative could be extremely important. Shepard reported back that Baker was interested in the station proposal and wanted additional information about it. In an August 24 letter, Beggs provided Baker "with a number of papers which will familiarize you with our argumentation, which present Mac Baldrige's strong views on the program, and which provide some information on the history and the current state-of-play of this endeavor." Beggs even sent Baker "a statement that might be used by the President in announcing a decision to go ahead with the Space Station."²

There was a report in early October that the White House was becoming "increasingly receptive toward approval of a NASA space station because of the political benefit such a decision could bring just prior to an election year." Reagan's possible 1984 opponent was former astronaut John Glenn (D-OH), who had endorsed the station, and Reagan's political advisers did not want the president's lack of support for the civilian space program to become an election issue, should Glenn be the Democrat candidate. The same report noted: "Another factor is Reagan's enthusiasm for the U.S. space program. Corporate officials who have discussed space with U.S. presidents since Dwight D. Eisenhower say Reagan is more openly enthusiastic about the program than any other, including John F. Kennedy."³

Another development acting in NASA's favor was that the space shuttle, at least in the public's eye, had been a spectacular success in its initial two years of flights. This success had demonstrated to the White House and Congress both the political benefits of highly visible human spaceflight activity and NASA's capability to bring a challenging development program to a successful outcome.

The question facing NASA and its White House supporters such as Gil Rye and Craig Fuller in late 1983 was how, given the national security community's opposition to the undertaking and its majority position within the SIG (Space), to get the space station decision before the president in a positive context. It was unlikely that Reagan would render a decision before hearing what his top advisers thought about the station. In addition, NASA first had to navigate its way through some confusing signals from the White House.

SHOULD NASA STAY THE COURSE?

In the final months of 1983, there were two developments that muddled the situation regarding the fate of NASA's space station proposal. They raised the issue of whether, on one hand, the space agency should accept a less ambitious compromise approach to its future programs or, on the other hand, abandon its current focus on space station advocacy and seek presidential approval for a longer-term space goal that would require the station for its achievement.

OMB Compromise Proposed

Although in its discussions with NASA with respect to the space station Office of Management and Budget (OMB) had continued throughout 1983 to oppose the project, the budget office internally was preparing itself to deal with

presidential approval of the undertaking. The FY 1985 and beyond that included in the FY 1985 budget process. The OMB permanent staff was getting the message that Ronald Reagan was likely to approve a way of showcasing U.S. technological leadership. At some point early in the year, the OMB Associate Director for Energy and the OMB space and science branch were going to happen. From OMB's perspective, the (Space) process were almost irrelevant, as it turned out correctly, the process, and OMB's assumption through ultimately turn out to be positive. The term budget of a positive station decision.

Reflecting its organizational skepticism about the FY 1985 budget process OMB's presidential decision to approve NASA's Beggs on October 19, OMB's budget "a space platform that would expand later on to include a manned mission." "we could afford to do something like to something like two billion dollar mission." Beggs's response to Khedouri was "no." "would not make such an offer unless NASA's proposal for a fully manned station."

Interpreting

NASA celebrated its 25th anniversary at the Air and Space Museum. Ronald Reagan on early space shuttle flights, make sense. Before the event NASA held out the possibility of including an indication of presidential approval. Not to be, Ed Meese was quoted as saying it would be in the shape of a space station.

Reagan in his remarks did indicate a broader concept of American exceptionalism except to galvanize our spirit as a people, to the potential." To NASA's dismay, much of that had been set forth by science and

We're not just concerned about the entire road, a "high road" if you will, and opportunity, that will spotlight the betterment of humankind. On that at NASA and the rest of America's space

Shepard reported back that Baker was wanted additional information about it. Baker "with a number of papers which tation, which present Mac Baldrige's which provide some information on the this endeavor." Beggs even sent Baker President in announcing a decision to

r that the White House was becoming of a NASA space station because of the bring just prior to an election year." former astronaut John Glenn (D-OH), gan's political advisers did not want the in space program to become an election t candidate. The same report noted: for the U.S. space program. Corporate with U.S. presidents since Dwight ly enthusiastic about the program than ³

IA's favor was that the space shuttle, at atacular success in its initial two years of o the White House and Congress both human spaceflight activity and NASA's ment program to a successful outcome. White House supporters such as Gil Rye given the national security community's ajority position within the SIG (Space), e the president in a positive context. It a decision before hearing what his top addition, NASA first had to navigate its om the White House.

TAY THE COURSE?

re two developments that muddled the space station proposal. They raised the e agency should accept a less ambitious grams or, on the other hand, abandon acy and seek presidential approval for a ire the station for its achievement.

mise Proposed

with respect to the space station Office had continued throughout 1983 to ternally was preparing itself to deal with

presidential approval of the undertaking, developing budget forecasts for Fiscal Year (FY) 1985 and beyond that included an allocation for the space station. The OMB permanent staff was getting "vibes" from its political leadership that Ronald Reagan was likely to approve some version of NASA's space station as a way of showcasing U.S. technological prowess and of maintaining space leadership. At some point early in the year Fred Khedouri, a political appointee as OMB Associate Director for Energy and Natural Resources, told the head of the OMB space and science branch, Jeff Struthers, "It's [the space station] going to happen." From OMB's perspective, the struggles during the SIG (Space) process were almost irrelevant. The actual space station decision, OMB believed, as it turned out correctly, would be made through the budget process, and OMB's assumption throughout 1983 was that the decision would ultimately turn out to be positive. Thus, OMB's goal was to limit the short-term budget of a positive station decision.⁴

Reflecting its organizational skepticism regarding the space station, early in the FY 1985 budget process OMB tried one maneuver to preempt a positive presidential decision to approve NASA's space station proposal. Meeting with NASA's Beggs on October 19, OMB's Khedouri offered to include in the NASA budget "a space platform that would initially be unmanned but that might be expanded later on to include a manned capability." Khedouri told Beggs that "we could afford to do something like this if the total runout cost could be kept to something like two billion dollars for the development of the platform." Beggs's response to Khedouri was "noncommittal"; Beggs judged that Khedouri "would not make such an offer unless he thought that there was a chance that NASA's proposal for a fully manned space station might succeed."⁵

Interpreting a Reagan Speech

NASA celebrated its 25th anniversary with an October 19 gala at the National Air and Space Museum. Ronald Reagan would attend, see a large-screen movie on early space shuttle flights, make some remarks, and cut an anniversary cake. Before the event NASA held out the slight hope that those remarks would include an indication of presidential support for the space station, but this was not to be. Ed Meese was quoted as saying, "NASA's birthday cake is not going to be in the shape of a space station."⁶

Reagan in his remarks did indicate how the space program fit into his broader concept of American exceptionalism, saying, "NASA's done so much to galvanize our spirit as a people, to reassure us of our greatness and of our potential." To NASA's dismay, much of his speech closely echoed the views that had been set forth by science adviser Keyworth several months earlier:

We're not just concerned about the next logical step in space. We're planning an entire road, a "high road" if you will, that will provide us a vision of limitless hope and opportunity, that will spotlight the incredible potential waiting to be used for the betterment of humankind. On this 25th anniversary, I would challenge you at NASA and the rest of America's space community: Let us aim for goals that will

carry us well into the next century. Let us demonstrate to friends and adversaries alike that America's mission in space will be a quest for mankind's highest aspiration—opportunity for individuals, cooperation among nations, and peace on Earth.

The first 25 years of NASA opened a new era. Let us all rededicate ourselves today that NASA's next 25 years will ensure that this new chapter in history will be an American era.⁷

Reagan's speech, which included a rather dismissive characterization of the space station as only a step in a broader space effort, seemed to pose a threat to NASA's strategy of focusing its advocacy on the station and the shorter-term missions it would serve. McCurdy comments that the speech "placed NASA executives in the curious position of having to assume that Reagan was not speaking for himself." NASA soon discovered that the speech was the product of Keyworth's Office of Science and Technology Policy (OSTP). It was not surprising, then, that the speech reflected Keyworth's view of the need for an expansive space effort. What was not clear was whether it also represented Ronald Reagan's views.⁸

NASA decided not to interpret Reagan's remarks as a signal of the president's desire to postpone a space station decision while a more ambitious vision for the future in space was developed. Beggs was quoted as asking, "Is the country ready for another Apollo-type effort? I would like to think so, but I doubt it ... We have to constrain those dreams to fit the kind of funding profile we are likely to get."⁹ Thus, NASA's focus remained on getting a positive decision on the space station before the end of the year. The long process of getting the space station issue formally before the president was soon to reach its climax.

FINDING THE PATH TO THE PRESIDENT

As the space station debate reached its final stage, there was a major change in the senior White House staff. National Security Adviser William Clark submitted his resignation on October 17, 1983; Clark had been engaged in a number of contentious battles with White House Chief of Staff James Baker and Secretary of State George Shultz, and had lost the confidence of senior adviser Michael Deaver and Ronald Reagan's wife Nancy, a behind-the-scenes powerful influence on the president. Reagan nominated Clark to replace James Watt as secretary of the interior, a position he assumed on November 18 after receiving Senate confirmation. Before Clark left the White House, there was a short-lived attempt to have Baker take Clark's job, with Deaver becoming chief of staff. Clark led the group within the White House that aborted this scheme.

Instead, Clark was replaced by his deputy, Robert "Bud" McFarlane, who had come from the State Department together with Clark in January 1982 and was a veteran of foreign policy and national security deliberations and actions. Unlike Clark, McFarlane was not a long-time Reagan associate and social friend, and his interaction with the president was on most days limited to the

15-minute national security briefing
tine. McFarlane would turn out to b
In his early weeks in office, as the
McFarlane was dealing with the after
a Korean airliner on September 1, k
October 23 attack on a Marine barr
people. He left dealing with the spac
larly Gil Rye.¹⁰

SIG (Space) in fall 1983 had also been created to oversee the commercialization of space technology. As that issue was approaching a presidential election, Rye had agreed to switch the forum through which SIG would be put before the president. Rather than deal with Ed Meese's agreement, to continue the work with the CCCT, with Ronald Reagan at the 1983 session to consider the lead agency for space, Rye explained that "Craig Fulmer will hereby SIG (Space) issues which are referred to the CCCT for resolution. The President would then be defin-

Rye and Fuller agreed on a similar issue. This switch had a certain logic: to advocate the space station as a civilian project, involving both civilian and national security, would require a Cabinet Council, according to Fuller's "recommendation." Unlike the situation in SIG, where the national security community were in the driver's seat, this would be counterbalanced by representatives of the civilian side. Fuller suggested that the Department of Interior would set the requirements for a space station, making the State Department's role "average over the final decision." In addition, the bill was part of the [space station] bill (plus the Air Force/NASA disputes" with respect to common sense."¹²

In scheduling the Cabinet Council "supported the space station concept of pros and cons regarding the und issue "to get lost due to internal riv national security community; the C process." At the end of an August 3 Reagan had indicated that he favor facility to promote space commerci modated in projected budgets. Ed M to be sure representatives of this p president was briefed on the station.

monstrate to friends and adversaries
 be a quest for mankind's highest
 eration among nations, and peace

era. Let us all rededicate ourselves
 hat this new chapter in history will

dismissive characterization of the
 effort, seemed to pose a threat to
 the station and the shorter-term
 its that the speech "placed NASA
 g to assume that Reagan was not
 d that the speech was the product
 ology Policy (OSTP). It was not
 eyworth's view of the need for an
 was whether it also represented

emarks as a signal of the president's
 ule a more ambitious vision for the
 quoted as asking, "Is the country
 I like to think so, but I doubt it ...
 kind of funding profile we are likely
 getting a positive decision on the
 long process of getting the space
 is soon to reach its climax.

THE PRESIDENT

stage, there was a major change in
 rity Adviser William Clark submit-
 ark had been engaged in a number

Chief of Staff James Baker and
 st the confidence of senior adviser
 nancy, a behind-the-scenes power-
 nated Clark to replace James Watt
 med on November 18 after receiv-
 ie White House, there was a short-
 b, with Deaver becoming chief of
 House that aborted this scheme.

y, Robert "Bud" McFarlane, who
 er with Clark in January 1982 and
 security deliberations and actions.
 time Reagan associate and social
 it was on most days limited to the

15-minute national security briefing that was part of the Reagan morning rou-
 tine. McFarlane would turn out to be a supporter of a strong space program.
 In his early weeks in office, as the station decision was coming to a head,
 McFarlane was dealing with the aftermath of the Soviet Union shooting down
 a Korean airliner on September 1, killing all 269 people aboard, and then an
 October 23 attack on a Marine barracks in Beirut, Lebanon, that killed 240
 people. He left dealing with the space station issue to his subordinates, particu-
 larly Gil Rye.¹⁰

SIG (Space) in fall 1983 had also been the venue for selecting the lead agency
 to oversee the commercialization of expendable launch vehicles (see Chap. 12).
 As that issue was approaching a presidential decision, Gil Rye and Craig Fuller
 had agreed to switch the forum through which the lead agency question would
 be put before the president. Rather than employ SIG (Space), they decided,
 with Ed Meese's agreement, to convene the Cabinet Council on Commerce
 and Trade (CCCT), with Ronald Reagan in attendance, for a November 16,
 1983, session to consider the lead agency issue. Commenting on this switch of
 venue, Rye explained that "Craig Fuller and I have worked out an arrangement
 whereby SIG (Space) issues which are principally civil or commercial in nature
 can be referred to the CCCT for resolution ... The final forum for presentation
 to the President would then be defined by the nature of the issue."¹¹

Rye and Fuller agreed on a similar switch with respect to the space station
 issue. This switch had a certain logic behind it, once Beggs had decided to
 advocate the space station as a civilian-only program rather than a facility serv-
 ing both civilian and national security purposes. In addition, the switch to the
 Cabinet Council, according to Fuller, "allowed a little bit broader participa-
 tion." Unlike the situation in SIG (Space), where the representatives of the
 national security community were in the majority, in the Cabinet Council they
 would be counterbalanced by representatives of domestic agencies. Rye later
 suggested that the Department of Defense (DOD), by stating that it had no
 requirements for a space station, made a "serious blunder," giving up its "lever-
 age over the final decision." In addition, DOD "paranoia over being stuck with
 part of the [space station] bill (plus a 'NASA-bashing' attitude) related to
 NASA/Air Force disputes" with respect to the space shuttle "overcame their
 common sense."¹²

In scheduling the Cabinet Council meeting, Fuller recognized that Reagan
 "supported the space station conceptually," but wanted to hear the full range
 of pros and cons regarding the undertaking. Fuller did not want the station
 issue "to get lost due to internal rivalry," particularly between NASA and the
 national security community; the CCCT venue provided "a little more open
 process." At the end of an August 3 lunch with a dozen aerospace executives,
 Reagan had indicated that he favored going ahead with a space station as a
 facility to promote space commercialization, if only its cost could be accom-
 modated in projected budgets. Ed Meese, who had attended the lunch, wanted
 to be sure representatives of this private sector interest were present as the
 president was briefed on the station, and thus readily agreed to the change of

venue. The chair of the CCCT, Secretary of Commerce Malcolm Baldrige, had developed a strong interest in the economic potentials of space activity; he was actively seeking to make his department the focal point for promoting that potential, and had also given his strong support to the space station idea. Given all of these factors, Rye and Fuller recognized that by presenting the station issue through the CCCT, the "right people" would be sitting around the table to present a positive picture to the president.¹³

Fuller and Rye visited NASA on November 18. Mark comments that by that meeting "Fuller was an open and strong supporter of the space station program" and "was confident that we would eventually be able to persuade the president to go ahead with it." One action Mark took following the meeting was to ask NASA's Langley Research Center to build for use at the presidential briefing a high-fidelity scale model of the station design NASA was advocating.¹⁴

On November 22, 1983, Fuller sent a memorandum to the members of the CCCT, announcing a December 1 meeting of the council to discuss "whether or not to proceed with the NASA development of a permanently based, manned Space Station." He attached to the memo Rye's August SIG (Space) 22-page report on the space station, noting that it contained "three options for the President's consideration," but that the "Secretary of Commerce recommends that the President approve both Space Station and enhanced Shuttle capability (i.e., both Options 1 and 2.)" This added an Option 4 for presidential consideration. Fuller in his memorandum noted that "at the SIG (Space) meeting on August 10, the majority of members supported either Option 1 or 3"; he omitted the fact that the majority had supported Option 3, deferring a decision. Fuller noted that "the meeting will consist of a summary of the options by Gil Rye ... and a presentation by NASA followed by a general discussion."¹⁵

One possible reason for scheduling the presidential briefing for December 1 was that on the prior day, November 30, Reagan was already scheduled to participate in a National Security Council meeting to discuss how to proceed with the Strategic Defense Initiative (SDI) that he had announced in eight months earlier. The trade weekly *Aviation Week & Space Technology* suggested that "an important element in the station decision could be which space-based missile defense development option is selected by President Reagan" at the November 30 meeting.¹⁶

Getting Ready

Terry Finn of the NASA Space Station Task Force was responsible for preparing the charts that Beggs would use in his briefing to the president. Rather than pull already-prepared charts from his large supply used in prior briefings, Finn decided to prepare all-new viewgraphs (transparent sheets of plastic upon which words and images could be printed). Finn in a 1985 interview noted that this presidential briefing "was very, very carefully constructed." It was "an effort to appeal to emotion and pride ... to adventure, to leadership, and portrays the Space Station in a very traditional sense of American greatness."

An unresolved issue was whether participation in the space station community with respect to such respect to the station project be- fer, NASA decided to separate the basic decision whether or not to that "we put together a couple o early editions of the briefing. An

Mark met with Rye on Nove The two discussed how to "deal v 1 briefing would "not be a decisio for the president." He predicted "zero out the Space Station," a c Rye's view that NASA would ha dent as part of the budget appe made in this arena." By schedulir "were going through the moti biters' so they could say they h events over the next two weeks

SELLING

The December 1 briefing was h commented that he, Mark, and House, but had "trouble getting not been alerted to the model's in, and we set it up and ... we pi graphs can be seen." The thre President will come in. And M meeting participants gathered, A model. Finn's assessment was th model "is worth ten thousand ..

Although the gathering had t actual attendance list made it clc net. Those present included Vice James Baker from Reagan's inne now-Secretary of the Interior Ju Attorney General William Frenc Deputy Secretary of Treasury I Paul Thayer, Undersecretary c Transportation James Burnley, McMahon, and William Niskar Among the dozen White Hous and his associate director Khedc

Commerce Malcolm Baldrige, the focal point for promoting that idea. Given that by presenting the station would be sitting around the table

18. Mark comments that by that time of the space station program "he was able to persuade the president following the meeting was to ask for use at the presidential briefing a NASA was advocating."¹⁴

Memorandum to the members of the staff of the council to discuss "whether or not of a permanently based, manned space station." The August SIG (Space) 22-page memorandum contained "three options for the future of Commerce recommends enhanced Shuttle capability Option 4 for presidential consideration: 'at the SIG (Space) meeting on either Option 1 or 3'; he omitted Option 3, deferring a decision.

a summary of the options by Gilchrist by a general discussion."¹⁵

Presidential briefing for December 1 was already scheduled to discuss how to proceed with the station. He had announced in eight months that the *Space Technology* suggested that "an alternative would be which space-based missile defense system Reagan" at the November

An unresolved issue was whether to mention the potential for international participation in the space station. Given the skepticism of the national security community with respect to such participation and in particular its concern with respect to the station project being a channel for unwanted technology transfer, NASA decided to separate the issue of international involvement from the basic decision whether or not to go ahead with the station at all. Finn recalled that "we put together a couple of charts on international [participation] in the early editions of the briefing. And they didn't make the final twelve."¹⁷

Mark met with Rye on November 22 to discuss the December 1 session. The two discussed how to "deal with OMB." Rye's view was that the December 1 briefing would "not be a decision meeting but rather an information meeting for the president." He predicted that in its passback to NASA, OMB would "zero out the Space Station," a decision that NASA would then appeal. It was Rye's view that NASA would have to take the space station issue to the president as part of the budget appeal process, and that "the real decision will be made in this arena." By scheduling the Cabinet Council meeting, he and Fuller "were going through the motions for the benefit of the nay-saying 'ankle-biters' so they could say they had their day in court."¹⁸ Rye was prescient; events over the next two weeks unfolded as he predicted.

SELLING THE SPACE STATION

The December 1 briefing was held in the White House Cabinet Room. Finn commented that he, Mark, and John Hodge took the model to the White House, but had "trouble getting in," possibly because the Secret Service had not been alerted to the model's arrival. Finn continued: "We finally [did] get in, and we set it up and ... we practiced with the lights to make sure the viewgraphs can be seen." The three placed the model "by the door where the President will come in. And Mark positions himself by the model." As the meeting participants gathered, Mark "would snag people and show them" the model. Finn's assessment was that if a "picture is worth a thousand words," a model "is worth ten thousand ... Whatever this model cost, it was worth it."¹⁹

Although the gathering had been convened as a meeting of the CCCT, the actual attendance list made it closer to a convening of the overall Reagan cabinet. Those present included Vice President George H.W. Bush, Ed Meese and James Baker from Reagan's inner circle, Secretary of Commerce Baldrige, the now-Secretary of the Interior Judge Clark, Secretary of Energy Donald Hodel, Attorney General William French Smith, Trade Representative William Brock, Deputy Secretary of Treasury R.T. McNamar, Deputy Secretary of Defense Paul Thayer, Undersecretary of Labor Barney Ford, Deputy Secretary of Transportation James Burnley, Deputy Director of Central Intelligence John McMahon, and William Niskanen from the Council of Economic Advisers. Among the dozen White House staff present were OMB director Stockman and his associate director Khedouri, science adviser Keyworth, Baker's deputy

Force was responsible for preparing the briefing to the president. Rather than simply used in prior briefings, Finn displayed sheets of plastic upon which Finn in a 1985 interview noted were carefully constructed." It was "an adventure, to leadership, and portrayal of American greatness."

Richard Darman, National Security Adviser McFarlane, Press Secretary Larry Speakes, and, of course, Fuller and Rye.

Although Gil Rye's space station report had been distributed in advance of the meeting, it is likely that few of the senior attendees had read it. Rye led off the session by summarizing the report and the options for presidential decision it had examined. Like Fuller in the memo calling the meeting, Rye did not mention that at the August 10 SIG (Space) meeting a majority had voted for Option 3, deferring a decision.

Beggs then presented NASA's 20-minute space station pitch. Hans Mark in his diary notes suggests that Beggs "unfortunately read from his notes too often, so that it [his briefing] did not seem to me to be as effective as it could have been." By contrast, another participant in the meeting described Beggs as "very eloquent. He stated it at just the right level, with vision, but not trying to commit to something too ambitious."²⁰

Beggs started off by quoting President John Kennedy: "The capacity to dominate space is essential to the United States as a leading world power." He then cited President Richard Nixon as he announced approval of the space shuttle, speaking of "man's epic voyage into space—a voyage the United States of America has led and still shall lead." Beggs noted that the focus on leadership had been "reaffirmed" in the Reagan administration's July 1982 National Space Policy. A next chart showed NASA's budget, peaking during the Apollo program and then, after a rapid decline, essentially flat at a much lower level since the last missions to the Moon. Then an image of the space shuttle was projected, with the claim that it represented "leadership in space for the 1980's." Beggs noted that the shuttle "flies beautifully" and "has captured worldwide attention," reassuring "our Allies of America's technological strength." Beggs transitioned to an image of the space station NASA was proposing, saying, "What's needed now, what was originally envisioned, is a place to shuttle to." Below the station image was written: "A highly visible symbol of U.S. strength." Beggs elaborated on the various functions that a space station would enable, suggesting that it would "dominate the space environment for twenty years" and "stake out some options for the future, enabling a President in the years to come to embark the United States upon missions that transcend the boundaries of earth." This comment was intended as a defense against Keyworth's suggestion that a decision on the station should wait for the decision on more expansive space goals. The next chart played to Ronald Reagan's interest in space commercialization, with Beggs noting that "space is going to become a place of business" and that "a space station is going to make that possible." It was followed by an image of astronauts in the shuttle payload bay, with Beggs suggesting that "some people say you can do it all in space with robots. In fact, you must have man. He—and she—are the essential ingredients." Playing to the central theme of the briefing, Beggs added: "The presence of man is the key to leadership in space."

Although NASA had decided to downplay competition with the Soviet Union as a theme in its case for a space station, the briefing contained two

images of the Soviet *Salyut* space station about "what the Soviets are up to. 1980s and the 1990s? Will they be so? The next chart listed the three SI proposed by Malcolm Baldrige, all with expanded capabilities. Beggs orbiter might be a "worthwhile" place in space." Deferring a space station commitments in space"; "sends a signal program is going to rest on its laurels that we are going to forfeit our hegemony graph in Beggs's presentation trace over the past quarter century. Beggs here in this room, we must look for ship in space for the next 25 years."

Reagan interrupted Beggs at one space station figured in the possibilities replied that a station was a necessary

Mark commented that Reagan "was more animated." As noted earlier Reagan had participated in a difficult review activity since his speech against ballistic missiles. In the intermediate options for how to proceed research and development as quickly make any commitment to deploying disappointed reaction to this recommendation should "hope and pray" that it would otherwise "we can expect nuclear meeting [the December 1 space station 'Star Wars.'" He added that "there sharp and quick on the uptake. He is in full command of the big picture."

After Beggs finished his presentation There are available two accounts from one prepared by Hans Mark, written Deputy Director of Central Intelligence complement one another.

Mark:

Keyworth says man's substantial manned presence: we catch up? Argue the moon right away goal. Wants to have study going to the

McFarlane, Press Secretary Larry

had been distributed in advance of the meeting. Rye led off the options for presidential decision. Calling the meeting, Rye did not expect a majority had voted for

space station pitch. Hans Mark unfortunately read from his notes too soon to be as effective as it could be in the meeting described Beggs at that level, with vision, but not trying

John Kennedy: "The capacity to remain as a leading world power." He announced approval of the space station—a voyage the United States government noted that the focus on leadership in space during the Apollo era, peaking during the Apollo era, essentially flat at a much lower level. An image of the space shuttle was used to "lead" the presentation. "Leadership in space for the future is beautiful" and "has captured the imaginations of America's technological leaders." The space station NASA was proposing was originally envisioned, is a place where various functions that a space station can perform, "dominate the space environment for the future, enabling a new era of United States upon missions that have never before. The next chart played to Ronald Reagan, with Beggs noting that "space is the future." "A space station is going to make it possible for astronauts in the shuttle payload to say you can do it all in space with the shuttle and the space station—are the essential ingredients." Briefing, Beggs added: "The president."

play competition with the Soviet Union, the briefing contained two

images of the Soviet *Salyut* space station. Beggs suggested that he was worried about "what the Soviets are up to. What are they planning to fly in the late 1980s and the 1990s? Will they be successful in their plans to dominate space?" The next chart listed the three SIG (Space) options plus the fourth option proposed by Malcolm Baldrige, approving both a space station and a shuttle with expanded capabilities. Beggs noted that an extended duration shuttle orbiter might be a "worthwhile" project, but "hardly America's next step in space." Deferring a space station decision, he said, "derails new commercial endeavors in space"; "sends a signal to the American people that their space program is going to rest on its laurels"; and "means that, in the years ahead, that we are going to forfeit our hard won leadership in space." A final viewgraph in Beggs's presentation traced the history of NASA's accomplishments over the past quarter century. Beggs closed his remarks by saying, "Now, today, here in this room, we must look forward ... The stakes are enormous: leadership in space for the next 25 years."²¹

Reagan interrupted Beggs at one point in his presentation, asking him how a space station figured in the possibility of establishing a base on the Moon. Beggs replied that a station was a necessary step on the way to the lunar surface.

Mark commented that Reagan "looked better than he did yesterday and he was more animated." As noted earlier, on the previous day, November 30, Reagan had participated in a difficult National Security Council meeting to review activity since his speech announcing his hope to develop a defense against ballistic missiles. In the interim, the SIG for Defense Policy had examined options for how to proceed and had recommended proceeding with research and development as quickly as technology would allow, but not to make any commitment to deploying a defensive system. Reagan's somewhat disappointed reaction to this recommendation was that the United States should "hope and pray" that it would be first to deploy a defensive system; otherwise "we can expect nuclear blackmail." Mark commented that "this meeting [the December 1 space station briefing] was more fun than the one on 'Star Wars.'" He added that "there is no doubt that the President is extremely sharp and quick on the uptake. He does not concern himself with details ... but he is in full command of the big picture."²²

After Beggs finished his presentation, the floor was opened for discussion. There are available two accounts from meeting participants of what transpired, one prepared by Hans Mark, writing in telegraphic style, and the other by Deputy Director of Central Intelligence John McMahon. The two accounts complement one another.

Mark:

Keyworth says man in space is not new ... Russians have substantial manned presence ... Russians have a station, why should we catch up? Argues that we should make the goal of going to the moon right away. Wants to optimize program toward that goal. Wants to have a "space summit" of the best people to study going to the moon. Wants to broaden outlook.

McMahon: George Keyworth commented that the MSS [manned space station] is not new and he feels we shouldn't emphasize a simple way station but rather go for a bold step and optimize for a goal which would mean something in space, advocating a moon space station. He readily admitted, however, that he didn't know what we would do on the moon or how we would get there but suggested over the next six months we study it to determine those answers.

Mark: Baldrige says that options are too narrow. Says we should include commercialization and try to put together a goal for the State of the Union message ... Baldrige commented that we have not heard yet from the private sector and that there is some talk that several companies may form a consortium to build the MSS and lease it back to the U.S. [government.] This generated a few smiles and smirks around the table.

McMahon: Baldrige then commented to the President that the options we considered were too narrow and probably we ought to undertake a study into the commercial and civil uses of space.

Mark: Thayer [says] man in space is attractive. DOD viewpoint is that there is no significant requirement for men in space. Concerned about effect that station would have on shuttle. Says it has to work. Says that shuttle is short of meeting operational commitment. Says cost of \$8B is optimistic. Will cost another \$18 to \$20B to get it into operation. Should not follow Russians. The Russians need men in space because they don't have the technology to do things automated. Says he cannot put it high on the priority list ... Says that if we had the money it would be a good thing to have. Real problem is that it will bleed off effort from the shuttle.

McMahon: Thayer then spoke of the impact that the MSS would have on diverting NASA's attention away from their problems with the shuttle and optimizing that for the purpose for which it was designed. He also felt that the MSS would not be a \$8 billion program but rather a \$20 billion one ... If we had the money maybe we could take the chance and do a thing like the MSS but there is a shortage of funds to do essential items and we can't visualize the MSS in any way recouping its investment. He said because of his own uncertainties on the future demand for dollars he proposed not scheduling the MSS now but deferring it for six months to a year.

Mark: Stockman concerned about cost. All other things being equal we ought to do this. Not unreasonable to spend \$10-\$12B per year [on NASA]. Says all things are not equal. Says we are bankrupt. Deficit is \$208B—unacceptable. Congress has not helped. Congress will not help and in an election year especially. Says this

is a crisis. \$600B de
wrong to start anyth
lion in the 1980's.
under control before
to \$30B before it i
nothing else. [Mark

McMahon: Stockman then spe
costs when resourc
least \$10 to \$15 bil
get to fracture. He
out perilously close

Mark: Mr. President, have
to do what they ou
Brock strongly supp
space. Must look at

McMahon: we must make inve
Bill Brock, U.S. Tr
that we really have

years because every
us 20 times over in
treasury. McNamar
has yet to see its fir
can get 20 to 1 fo
[Secretary of the
Treasury and set up
an emotional talk c
we never see them
gave the Indians w
fold was eaten inste

Mark: Fuller [comments
more. Makes the f
are important. Par
build platform ... V
the opportunity to

McMahon: Craig Fuller then
commercial interes
feels that if the Pr
industry is with hin
partners cheering u
all withdrew.

As the meeting broke up, accord
Clark, Thayer and others came by
During the meeting, Ronald F
make a decision today, but just wa

1 that the MSS [manned space station] we shouldn't emphasize a simple or a bold step and optimize for a winning in space, advocating a moon mission, however, that he didn't see the moon or how we would get there in the next six months we study it to

re too narrow. Says we should try to put together a goal for the ... Baldrige commented that the private sector and that there is a consortium may form a consortium to go to the U.S. [government.] This is a problem around the table.

he President that the options we need probably we ought to understand the military and civil uses of space.

attractive. DOD viewpoint is that it is a problem for men in space. Concerned about the shuttle. Says it has to be able to meet operational commitments. Will cost another \$18 to \$20 billion. Should not follow Russians. The reason they don't have the technology. Says he cannot put it high on the list. Says if we had the money it would be a problem is that it will bleed off effort

act that the MSS would have on the way from their problems with the shuttle or the purpose for which it was built. The MSS would not be a \$8 billion mission one ... If we had the money now and do a thing like the MSS we could do essential items and we could be recouping its investment. He has doubts on the future demand for building the MSS now but deferring

st. All other things being equal we are unable to spend \$10-\$12B per year. The cost is not equal. Says we are bankrupt. The cost is not equal. Congress has not helped in an election year especially. Says this

McMahon:

Mark:

McMahon:

Mark:

McMahon:

is a crisis. \$600B deficits in 3 years. Says that the environment is wrong to start anything. We will run up a deficit of over one trillion in the 1980's. Must win election first and get Congress under control before we make commitment that will cost \$20B to \$30B before it is finished. Must have defense buildup but nothing else. [Mark commented: "an eloquent speech."]

Stockman then spoke, saying that the MSS was excessive in costs when resources are scarce. He felt that it would cost at least \$10 to \$15 billion and that amount would push the budget to fracture. He noted that the \$3.5 trillion program puts us out perilously close to fiscal bankruptcy ... He said that you, Mr. President, have got to get hold of Congress and force them to do what they ought to do.

Brock strongly supports the economic argument for going into space. Must look at the quality of govt. expenditures. Says that we must make investments.

Bill Brock, U.S. Trade Representative, then spoke. He claimed that we really haven't spent any dollars [in space] in the past 20 years because every dollar spent in space has been returned to us 20 times over in economic growth and flow of dollars to the treasury. McNamar interrupted him by saying that the Treasury has yet to see its first dollar from space and if it's true that we can get 20 to 1 for an investment like the MSS then he and [Secretary of the Treasury] Don Regan are going to leave Treasury and set up a company to do that ... Brock then gave an emotional talk on how dollars spent on welfare are wasted, we never see them and even noted how the seed corn that we gave the Indians which would have increased their harvest tenfold was eaten instead of planted.

Fuller [comments that] process is finished—don't need any more. Makes the private sector case. Definition and certainty are important. Partners want certainty. Private sector will not build platform ... We are not talking about much money. Have the opportunity to make an investment that we want to make. Craig Fuller then commented about he must represent U.S. commercial interests with whom he has been in touch and he feels that if the President makes this commitment that U.S. industry is with him and that he would even have our European partners cheering us on. Whereupon the meeting ended and we all withdrew.

As the meeting broke up, according to Hans Mark, "the President, Smith, Clark, Thayer and others came by to look at the model."²³

During the meeting, Ronald Reagan commented that "he didn't plan to make a decision today, but just wanted to hear everyone out." There had been

during the meeting one light-hearted exchange, also typical of how Reagan dealt with a contentious issue. As Treasury's McNamar was protesting that there had been no economic return on space spending, Attorney General William French Smith interrupted him, saying, "I'll bet the Comptroller of Ferdinand and Isabella made exactly the same speech when Columbus proposed to them that he sail west to reach the far east." Reagan quickly reacted, commenting, "Yes, Bill, you're right, but you remember that the Comptroller won the argument: Isabella had to pawn her jewels to pay for Columbus' trip. What I want to know is, who has the jewels around here?" According to Mark, "There was general laughter at this quip and from then on, there was definitely a more relaxed atmosphere in the room."

Mark "was disappointed at the outcome of the meeting ... there were more people against us than for us." Mark told Beggs: "I thought we did not lose anything at the meeting but that we also did not gain much." Beggs replied: "You're wrong. I could tell the president was with us. He winked at me a couple of times while the other people were making their critical remarks."²⁴

Beggs was correct. In his diary entry for December 1, Ronald Reagan wrote: "Cabinet Council on 'space' and where we go. The issue is whether to move on a program for a permanent manned space station. I'm for it as I think most everyone is but the question is funding such a new course in face of our deficits."²⁵ The space station decision had come down to whether the funding for the program could be accommodated in an acceptable overall NASA budget, not only for FY 1985, but for the following years of what Reagan and his associates hoped would be a second term in the White House.

DECISION MADE, PROTESTED, REAFFIRMED

There were in the first Reagan term two paths to presidential approval of a policy initiative. If the initiative required a policy decision with minimal budget implications, it most often would reach the president through one of the Cabinet Councils or, in the case of national security issues, an SIG or the National Security Council itself. If the initiative had significant budget as well as policy implications, the path to president's desk usually ran through the annual budget process. Although SIG (Space) had spent the better part of a year debating the space station issue, it would be through this second path that Ronald Reagan would give final approval to the space station program. This was not a surprise. Both NASA and those handling the station policy issue in the White House, particularly Gil Rye and Craig Fuller, had for several months been saying that the final decision on whether to proceed with the station would come as the FY 1985 budget for NASA was finalized.

Space Station Approved

As NASA submitted its FY 1985 budget request to OMB in September 1983, it had asked for \$235 million to support space station Phase B detailed design efforts, and forecast that it would need an additional \$460 million in the next

two fiscal years to complete that phase. The Phase B budget game, did not include a Phase C space station development efforts, even though that phase was to begin operation in 1991. NASA estimated that over \$6 billion in development would be needed in FY 1989, the last Reagan administration; this was a need that NASA reflected in its budget request.

A crucial step in the annual "Review," at which the OMB presented the agency's budget request to the President, was the OMB's recommendations for the OMB's review. The ship makes an initial decision on whether to approve it to the agency in a "pass" presentation with respect to the N "the need for increased funding in with the desire of NASA to initiate major R&D programs leads to a major over 1984 and to dramatic growth were: "Should a commitment be approved for 1985? If not, what are the constraints for related activities?" The seven-page "issue paper" on the NASA's budget proposal was on the space station costing \$8 billion, and a "fully developed Space Station" at a billion figure "does not include a and applications to be performed. of NASA's cost estimates, since ' costs for large projects are a factor uncertainty as to whether a Space meeting the future needs of most considering international participation. "NASA may in fact give away our participation. Another consideration natives exist on two fronts: extensive space platforms; and the President initiative, which may require considerable large unmanned platforms."

Given all these considerations, "commitment" but "consider establishing space facilities." If this recommendation a small amount of money to explore for consideration in future budget

nge, also typical of how Reagan's McNamara was protesting that excessive spending, Attorney General Regan, "I'll bet the Comptroller of the Treasury said that in his speech when Columbus professed to go far east." Reagan quickly reacted, "I remember that the Comptroller said that the jewels to pay for Columbus' trip were round here?" According to Mark, from then on, there was definitely

from the meeting ... there were more eggs: "I thought we did not lose and I did not gain much." Beggs replied: "I was with us. He winked at me as he was making their critical remarks."²⁴ On December 1, Ronald Reagan wrote: "On 30. The issue is whether to move the station. I'm for it as I think most of us are. A new course in face of our defeat is not a new course in face of our defeat. It is a question of whether the funding for the station is acceptable overall NASA budget, and the years of what Reagan and his associates at the White House.

DECISION, REAFFIRMED

months to presidential approval of a policy decision with minimal budget review by the president through one of the major security issues, an SIG or the like. The White House had significant budget as well as the president's desk usually ran through the White House. The White House had spent the better part of a year through this second path that led to the space station program. This handling of the station policy issue in the Reagan White House, had for several months decided whether to proceed with the station program. A decision was finalized.

proved

request to OMB in September 1983, the space station Phase B detailed design cost an additional \$460 million in the next

two fiscal years to complete that phase of the program. NASA, playing a bit of a budget game, did not include any money in its five-year budget forecast for the Phase C space station development phase that would follow the design efforts, even though that phase would have to start in FY 1987 if the station were to begin operation in 1991, as NASA was promising. The OMB staff estimated that over \$6 billion in development funding would be needed by FY 1989, the last Reagan administration budget request if the president were reelected; this was a need that NASA hoped to downplay by not mentioning it in its budget request.

A crucial step in the annual budgeting process is the OMB "Director's Review," at which the OMB professional staff presents its evaluation of an agency's budget request to the OMB director and his top assistants, with its recommendations for the OMB position. After this review, the OMB leadership makes an initial decision on an agency budget allowance and communicates it to the agency in a "passback." In its November Director's Review presentation with respect to the NASA budget, the OMB staff pointed out that "the need for increased funding in 1985 to meet approved commitments along with the desire of NASA to initiate a Space Station and expand sharply all of its major R&D programs leads to a requested \$0.9 billion increase [in FY 1985] over 1984 and to dramatic growth in outyear costs." Key questions for decision were: "Should a commitment to an initial manned Space Station be approved for 1985? If not, what are the appropriate funding levels and constraints for related activities?" The material for the Director's Review included a seven-page "issue paper" on the space station. That paper pointed out that NASA's budget proposal was only for an "initial stripped down version" of a space station costing \$8 billion, and that NASA acknowledged that the cost of "a fully developed Space Station may run as high as \$28 billion." Also, the \$8 billion figure "does not include any operating costs ... or the cost of science and applications to be performed on the Space Station." The staff was skeptical of NASA's cost estimates, since "history shows that NASA estimates of total costs for large projects are a factor of 2-10 too low"; also, there was "major uncertainty as to whether a Space Station would be the most suitable means for meeting the future needs of most space user groups." Noting that NASA was considering international participation in the station, the staff suggested that "NASA may in fact give away our technological leadership" through such participation. Another consideration was that "other possibly higher priority alternatives exist on two fronts: extended duration shuttle missions and unmanned space platforms; and the President's DABM [Defense against Ballistic Missiles] initiative, which may require construction of heavy lift launch vehicles and large unmanned platforms."

Given all these considerations, the staff recommendation was to "defer commitment" but "consider establishing a long term goal to establish permanent space facilities." If this recommendation were accepted, OMB would "provide a small amount of money to explore alternatives to the manned Space Station for consideration in future budgets."²⁶

This recommendation was accepted by OMB leadership, and in its passback to NASA, OMB included only \$6 million for continued low-level space studies. This was essentially a holding decision; OMB knew that NASA would protest the low budget allocation, and that the actual decision on whether or not to approve the space station would be made either by the president's senior advisers or by Ronald Reagan himself in the final stages of formulating the NASA FY 1985 budget.

NASA in late November did indeed protest the OMB budget offer, beginning the appeal process that would eventually reach the president. The final stop on that path was the White House Budget Review Board, consisting of James Baker, Ed Meese, and David Stockman. The board meeting on the NASA FY 1985 budget was held on December 2, the day after the station briefing to the president. Beggs made his presentation on the overall NASA budget to the three top White House aides, and there was a lengthy discussion of whether NASA's cost estimates for its core program (without the space station) were accurate. But NASA's budget request for the space station, by now reduced through negotiations between NASA and OMB to \$150 million, was not addressed; rather, NASA was informed that the space station decision "would require a special meeting with the president before a final decision could be made." That meeting was scheduled for Monday, December 5.

On December 3, Meese, Baker, and Stockman sent a decision memorandum to the president that outlined four alternatives for the NASA budget.

- Alternative I would fund NASA for FY 1985-1989 at the original target that OMB had provided some months earlier, curtailing some ongoing projects and not allowing initiation of any new projects during the next five years;
- Alternative II would provide an increase above the original target sufficient to maintain all current projects and to allow some new starts other than the space station; this alternative would add \$4.97 billion to the NASA five-year budget;
- Alternative III would augment the budget proposed in alternative II by funding the full NASA request for the space station; this alternative would add \$12.2 billion to the NASA five-year budget;
- Alternative IV would provide NASA with a fixed five-year funding allowance that would yield 1 percent real growth and allow a start on the space station; this alternative would add \$10.4 billion to the NASA budget.

Meese, Baker, and Stockman advised the president to "approve Alternative II if you decide to defer approval of the Space Station" and to "approve Alternative IV if you decide to give your approval to starting the Space Station in FY1985."²⁷

NASA in fact did not have to wait for the outcome of the December 5 meeting to learn that Reagan had approved the space station and thus that funds for the program would be included in its FY 1985 budget. In response to the memo from the members of the Budget Review Board, Reagan indicated that he would

approve the station, but asked them not to communicate it to the NA Khedouri on Saturday, December 3; he requested would be in the president's office the following Monday (December 5), the president." Rye comments that "Beggs would learn of the Space Station 'ankle-biters'". Craig [Fuller] and I were

NASA's meeting with Reagan was scheduled for December 5. It took place in a very private setting. Just before that meeting, Reagan engaged in a three-way televised conversation with Kohl and astronauts aboard space shuttle launched on November 28 for a ten-day mission to carry in its payload bay the Space Shuttle contribution to the shuttle program. Spacelab development, and aboard the shuttle, scientist Ulf Merbold. (In addition to fly on a NASA mission.) In his remarks, he noted: "This mission is also a shining example of its best ... Building on that good vision, another country has joined one of our nations. It is an exciting first."²⁹

Going almost directly from his office, the space station certainly reinforced the president's support for the space program. At the Cabinet meeting, Fuller "sat in Beggs' corner," while the budget director tried to get NASA to accept an increase in NASA's overall budget. NASA's budget had suffered over the years, and an increase was needed. Stockman, in the December 3 decision memorandum, noted that Beggs accepted that offer, that the space station decision was approved, and that Reagan noted: "A Budget appeal meeting OK there & can still start to plan a space station. The December 5 meeting not to come to announce it himself, and already announced."³⁰

Decision

With Reagan's directive not to publicize the decision, other senior members of his administration still had time to weigh in against suc-

y OMB leadership, and in its passback for continued low-level space studies. OMB knew that NASA would protest actual decision on whether or not to either by the president's senior advisors. The final stages of formulating the NASA

protest the OMB budget offer, beginning to reach the president. The final Budget Review Board, consisting of Stockman. The board meeting on the December 2, the day after the station was presented on the overall NASA issues, and there was a lengthy discussion of the core program (without the space station request for the space station, by now NASA and OMB to \$150 million, was decided that the space station decision would be presented to the president before a final decision scheduled for Monday, December 5. Stockman sent a decision memorandum alternatives for the NASA budget.

for FY 1985-1989 at the original target months earlier, curtailing some ongoing and of any new projects during the next

increase above the original target sufficient to allow some new starts other alternative would add \$4.97 billion to the

the budget proposed in alternative II by the space station; this alternative would be a five-year budget; NASA with a fixed five-year funding allowed growth and allow a start on the space \$10.4 billion to the NASA budget.

the president to "approve Alternative II for the Space Station" and to "approve Alternative III for starting the Space Station in FY1985."²⁷ the outcome of the December 5 meeting on the space station and thus that funds for the 1985 budget. In response to the memo from the Board, Reagan indicated that he would

approve the station, but asked them not to discuss that decision, with the exception of communicating it to the NASA leadership. Beggs met with OMB's Khedouri on Saturday, December 3; he was told that "the \$150 million we had requested would be in the president's budget and that this would be confirmed the following Monday (December 5, 1983) at a formal decision meeting with the president." Rye comments that "it must be one of the great ironies ... that Beggs would learn of the Space Station decision from Fred Khedouri (one of the 'ankle-biters'). Craig [Fuller] and I were sworn to secrecy."²⁸

NASA's meeting with Reagan was scheduled for 11:00 a.m. on the morning of December 5. It took place in a very positive context from NASA's perspective. Just before that meeting, Reagan, speaking from the Oval Office, had engaged in a three-way televised conversation with German Chancellor Helmut Kohl and astronauts aboard space shuttle *Columbia*. The shuttle had been launched on November 28 for a ten-day mission; it was the first shuttle flight to carry in its payload bay the Spacelab pressurized laboratory, a European contribution to the shuttle program. Germany had been the lead funder of Spacelab development, and aboard the mission was the first German to fly on the shuttle, scientist Ulf Merbold. (In fact, Merbold was the first non-American to fly on a NASA mission.) In his remarks during the conversation, Reagan noted: "This mission is also a shining example of international cooperation at its best ... Building on that good will, this is the first time a citizen from another country has joined one of our space missions as a member of the crew. It is an exciting first."²⁹

Going almost directly from his orbital conversation to his meeting on the space station certainly reinforced the president's positive attitude toward NASA and the space program. At the Cabinet Room meeting (Fig. 9.1), Meese, Rye, and Fuller "sat in Beggs corner," while Khedouri backed up Stockman. The budget director tried to get NASA to agree to develop the space station without an increase in NASA's overall budget in future years. Beggs insisted that NASA's budget had suffered over the previous decade, and that an "appropriate increase" was needed. Stockman finally was willing to offer Alternative IV in the December 3 decision memorandum, a 1 percent annual increase above inflation. Beggs accepted that offer, and Ronald Reagan said: "Done." With that, the space station decision was made. Writing in his diary that evening, Reagan noted: "A Budget appeal meeting on N.A.S.A.'s request. I think we're OK there & can still start to plan a space station." Reagan instructed those in the December 5 meeting not to communicate his choice to others; he wanted to announce it himself, and already had a plan for how to make that announcement.³⁰

Decision Protested

With Reagan's directive not to publicize his positive decision on the space station, other senior members of his administration apparently believed that they still had time to weigh in against such a choice. On December 9, Director of



Fig. 9.1 Ronald Reagan gave final approval of NASA's space station program at a December 5, 1983, budget review meeting in the White House Cabinet Room. Sitting to the president's right is Edwin Meese. Across the table (l to r) are OMB Director David Stockman, Secretary of the Treasury Donald Regan, an unidentified person, NASA Administrator James Beggs, Cabinet Secretary Craig Fuller (standing), Gil Rye (against wall), NASA Comptroller C. "Tommy" Newman, and Deputy National Security Adviser Admiral John Poindexter. (Photograph courtesy of Reagan Presidential Library)

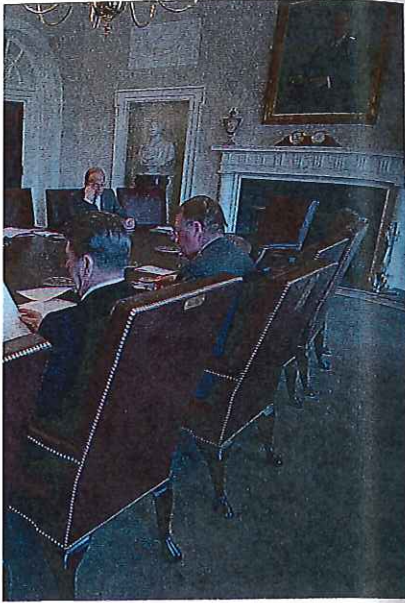
Central Intelligence William Casey, a long-time Reagan adviser, wrote to Meese, noting the lack of agreement at the December 1 meeting and saying, "I believe it is evident to everyone that without a national consensus it is questionable whether or not the nation can maintain the high level of commitment necessary to sustain, over the long run, major parallel efforts such as strategic forces and intelligence capabilities modernization, defense against ballistic missiles, and a bold new civil space initiative." Casey said that he wanted to "continue to stress that before the U.S. embarks on major new space initiatives we owe the President the complete menu of options which will permit him to address the civil program within the context of our national security needs."³¹

In an undated memorandum to Meese, possibly coordinated with the communication from Casey, Secretary of Defense Caspar Weinberger reiterated his opposition to going ahead with the station. He noted that the December 1 Cabinet Council meeting "underscored the fact that there is little support in government, the scientific community, or the private sector for NASA's

proposal to initiate development of a manned space station. The US commitment to a manned space station is a major engineering effort rather than an incremental expansion of expertise and resources from the task force. The station would follow along lines more likely to produce economic return. He thought that "one should not create a strained environment with any initiative that promises no return. The manned space station does not support the thrust of Jay Keyworth's effort to assemble the best talent in the industry to chart bold new directions for the civilian space program. It would provide the President with a context of national security and space exploration." Weinberger told Meese that he would support the President's commitment to the NASA space station program.

Concerned that these protests might undermine the December 5 decision in favor of the space station, the President's closure to the SIG (Space) process regarding the station was a "Memorandum of Decision" for the station's development. That three-page document will establish a permanent manned presence in space. "We will establish a permanent manned presence in space. Rye also drafted the memo by Meese and McFarlane, forwarding the memo said: "While we certainly understand the growing deficits and Paul Thayer's assessment of the Cabinet Room meeting] that there are no alternatives to the Space Station, we believe that the interest in your immediate approval of the program is unnecessary and probably politically motivated. The goal of returning to the moon ... You would preserve the option for more a manned space station perhaps during your second term when the memo concluded by recommending 'during your State of the Union Address signed the memorandum, but Meese did not sign it.'³³

The strong protests by Casey and Weinberger, did not deter Ronald Reagan. He announced the space station, and by early December he announced that decision. It was thus a major funding for space station design in the important, the upcoming high-profile commitment by the president himself that his



1 of NASA's space station program at a the White House Cabinet Room. Sitting ss the table (l to r) are OMB Director Donald Regan, an unidentified person, ecretary Craig Fuller (standing), Gil Rye nmy" Newman, and Deputy National otograph courtesy of Reagan Presidential

ong-time Reagan adviser, wrote to he December 1 meeting and saying, hout a national consensus it is ques-aintain the high level of commitment ajor parallel efforts such as strategic ernization, defense against ballistic ative." Casey said that he wanted to embarks on major new space initia-: menu of options which will permit the context of our national security

, possibly coordinated with the com-ense Caspar Weinberger reiterated his ion. He noted that the December 1 the fact that there is little support in or the private sector for NASA's

proposal to initiate development of a manned space station." He argued that "US commitment to a manned space station would be better characterized as a major engineering effort rather than a bold new space initiative" and that "undertaking an engineering project of this magnitude would divert NASA expertise and resources from the task of pressing the frontiers of technology along lines more likely to produce economic and scientific benefits for the nation." He thought that "one should not further stress our resource constrained environment with any initiative that does not promise extraordinary return. The manned space station does not meet this criterion." Weinberger supported "the thrust of Jay Keyworth's proposal for a 'space summit' to assemble the best talent in the industrial and scientific communities and to chart bold new directions for the civil space program," since such a gathering "would provide the President with a range of options to weigh in the dual context of national security and space leadership." Given these arguments, Weinberger told Meese that he would "strongly recommend deferring commitment to the NASA space station proposal."³²

Concerned that these protests might cause President Reagan to reverse his December 5 decision in favor of the space station, and likely to give a sense of closure to the SIG (Space) process regarding the space station, Gil Rye drafted a "Memorandum of Decision" for the president to sign recording his approval of station development. That three-page memo stated that "the United States will establish a permanent manned presence in space through development of a manned Space Station." Rye also drafted a cover memorandum to be signed by Meese and McFarlane, forwarding the decision memo to Reagan. The cover memo said: "While we certainly understand Dave Stockman's concern about growing deficits and Paul Thayer's assessment [expressed at the December 1 Cabinet Room meeting] that there are no current military requirements for a Space Station, we believe that the interests of the nation would be best served with your immediate approval of the program." The memo added: "We believe it is unnecessary and probably politically unwise for you to commit now to a goal of returning to the moon ... Your approval of a Space Station program would preserve the option for more ambitious goals sometime in the future, perhaps during your second term when deficits are under better control." The memo concluded by recommending "public announcement of your decision during your State of the Union Address." National Security Adviser McFarlane signed the memorandum, but Meese did not, and thus it was not forwarded to the president.³³

The strong protests by Casey and Weinberger, two of his most senior advisers, did not deter Ronald Reagan. He had made his decision to approve the space station, and by early December preparations were already underway to announce that decision. It was thus through the combination of inserting funding for space station design in the NASA FY 1985 budget and, more important, the upcoming high-profile announcement of space station development by the president himself that his approval would become public.

NOTES

1. Hans Mark, *The Space Station: A Personal Journey* (Durham, NC: Duke University Press, 1987), 179–180.
2. Letter from James Beggs to James A. Baker, III, August 24, 1983, Folder 12776, NHRC.
3. Craig Covault, "Defense Dept. Studies Aid Space Station," *AWST*, October 3, 1983, 19.
4. Interviews with Jeff Struthers, May 29, 2016, and Norine Noonan, September 23, 2016. Noonan in 1983 was an OMB budget examiner working under Struthers.
5. Mark, *The Space Station*, 181–182.
6. Meese's quote is in Michael A. G. Michaud, *Reaching for the High Frontier: The American Pro-Space Movement, 1972–1984* (New York: Praeger Publishers, 1986), 281.
7. Ronald Reagan: "Remarks at the 25th Anniversary Celebration of the National Aeronautics and Space Administration," October 19, 1983. Online by Gerhard Peters and John T. Woolley, APP, <http://www.presidency.ucsb.edu/ws/?pid=40662>.
8. Howard McCurdy, *The Space Station Decision: Incremental Politics and Technological Change* (Baltimore, MD: The Johns Hopkins Press, 1990), 181.
9. Craig Covault, "NASA Answers Planning Challenge," *AWST*, October 31, 1983, 22.
10. For an evaluation of Clark's tenure as national security adviser, see John P. Burke, *Honest Broker? The National Security Adviser and Presidential Decision Making* (College Station, TX: Texas A&M Press, 2009).
11. Memorandum from Gil Rye to Robert McFarlane, "Membership on the Senior Interagency Group for Space," November 30, 1983, Files of Valerie Neal.
12. Letter from Gil Rye to Hans Mark commenting on a draft of Mark's *The Space Station* book, October 25, 1984, cited with Rye's permission.
13. Interviews with Craig Fuller, December 14, 2016; with Edwin Meese, October 12, 2016; and Gil Rye, May 20, 2016.
14. Mark, *The Space Station*, 184.
15. Cabinet Affairs Staffing Memorandum, "CCCT Meeting on the U.S. Space Station," November 22, 1983, with attached memorandum from Craig Fuller to Members of the Cabinet Council on Commerce and Trade, "Space Station," undated, CIA-RDP85M00363R001002300012-7, CREST.
16. "Washington Roundup," *Aviation Week & Space Technology*, November 28, 1983, 17.
17. Interviews of Terence Finn by Sylvia Fries, June 12 and July 10, 1985, NHRC.
18. Handwritten note from Hans Mark to James Beggs, "In Strictest Confidence," November 23, 1983. I am grateful to Dr. Mark for providing me a copy of this note. Rye letter to Hans Mark, October 25, 1984.
19. Interview of Terence Finn by Sylvia Fries, June 12, 1985, NHRC.
20. Hans Mark, "Diary Notes for December 1, 1983," provided to the author by Dr. Mark; M. Mitchell Waldrop, "The Selling of the Space Station," *Science*, February 24, 1984.
21. NASA, "Revised Talking Points for the Space Station Presentation to the President and the Cabinet Council," with attached presentation charts, November 30, 1983, File 12766, NHRC.
22. Hans Mark, "Diary Notes for Dec meeting on the Strategic Defense Ini ganfiles.com/19831130-nsc-96-sdi.
23. This account of the December 1 m "White House Meeting," dated 1 Memorandum for the Record, "Su Commerce Meeting," Decembe 300430043-0, CREST. Dr. Mark pi
24. Mark, *The Space Station*, 186–187.
25. Hans Mark, "Diary Notes for Dece *Reagan Diaries* (New York: Harper
26. Director's Review material is from O Aeronautics and Space Administrat Station sections, undated but Nove
27. Memorandum from Edwin Mees A. Stockman to the President, " December 3, 1983, Box 9, Agency F Council, RRL.
28. Hans Mark, "Diary Notes for Decen Mark. Mark, *The Space Station*, 190.
29. Ronald Reagan: "Remarks During : Kohl of the Federal Republic of t Shuttle Columbia," December 5, : <http://www.presidency.ucsb.edu/v>
30. McCurdy, *The Space Station Decision*
31. Memorandum from William Casey t of 1 December 1983," Decen 1901730053-2, CREST.
32. Memorandum, unsigned but from "Manned Space Station," undated CREST.
33. Memorandum of Decision, "Develo memorandum from Edwin Meese : "Space Station Decision," Decen Khedouri to Craig Fuller, "Space S 12, 1983, all in Box 7, Outer Space

ES

Personal Journey (Durham, NC: Duke

A. Baker, III, August 24, 1983, Folder

ies Aid Space Station," *AWST*, October 3,

9, 2016, and Norine Noonan, September
1 OMB budget examiner working under

chaud, *Reaching for the High Frontier: The
72-1984* (New York: Prager Publishers,

h Anniversary Celebration of the National
n," October 19, 1983. Online by Gerhard
?, <http://www.presidency.ucsb.edu/ws/>

tion Decision: Incremental Politics and
D: The Johns Hopkins Press, 1990), 181.
anning Challenge," *AWST*, October 31,

national security adviser, see John P. Burke,
Adviser and Presidential Decision Making
ress, 2009).

rt McFarlane, "Membership on the Senior
mber 30, 1983, Files of Valerie Neal.
ommenting on a draft of Mark's *The Space*
ed with Rye's permission.
ber 14, 2016; with Edwin Meese, October
6.

um, "CCCT Meeting on the U.S. Space
attached memorandum from Craig Fuller
on Commerce and Trade, "Space Station,"
002300012-7, CREST.

Week & Space Technology, November 28,

Fries, June 12 and July 10, 1985, NHRC.
to James Beggs, "In Strictest Confidence,"
o Dr. Mark for providing me a copy of this
ber 25, 1984.

Fries, June 12, 1985, NHRC.
mber 1, 1983," provided to the author by
The Selling of the Space Station," *Science*,

or the Space Station Presentation to the
cil," with attached presentation charts,
HRC.

22. Hans Mark, "Diary Notes for December 1." Minutes of the November 30 meeting on the Strategic Defense Initiative can be found at <http://www.thereaganfiles.com/19831130-nsc-96-sdi.pdf>.
23. This account of the December 1 meeting is based on Hans Mark's notes on "White House Meeting," dated December 1, 1983, and John McMahon, Memorandum for the Record, "Summary of Cabinet Council on Trade and Commerce Meeting," December 1, 1983, CIA-RDP86B00885R000300430043-0, CREST. Dr. Mark provided a copy of his notes to the author.
24. Mark, *The Space Station*, 186-187.
25. Hans Mark, "Diary Notes for December 1, 1983"; Douglas Brinkley, ed. *The Reagan Diaries* (New York: HarperCollins, 2007), 201.
26. Director's Review material is from Office of Management and Budget, "National Aeronautics and Space Administration, 1985 Budget," Overview and Space Station sections, undated but November 1983.
27. Memorandum from Edwin Meese III, James A. Baker III, and David A. Stockman to the President, "NASA Budget for FY 1985-FY 1989," December 3, 1983, Box 9, Agency Files, Executive Secretariat, National Security Council, RRL.
28. Hans Mark, "Diary Notes for December 2, 1983," provided to the author by Dr. Mark. Mark, *The Space Station*, 190. Rye letter to Hans Mark, October 25, 1984.
29. Ronald Reagan: "Remarks During a Conference Call With Chancellor Helmut Kohl of the Federal Republic of Germany and Crewmembers of the Space Shuttle Columbia," December 5, 1983. Online by Peters and Woolley, APP, <http://www.presidency.ucsb.edu/ws/?pid=40834>.
30. McCurdy, *The Space Station Decision*, 185; Brinkley, ed. *The Reagan Diaries*, 295.
31. Memorandum from William Casey to Edwin Meese, "Cabinet Council Meeting of 1 December 1983," December 9, 1983, CIA-RDP92B00181R001901730053-2, CREST.
32. Memorandum, unsigned but from Caspar Weinberger, to Edwin Meese, "Manned Space Station," undated, CIA-RDP92B00181R001901730069-5, CREST.
33. Memorandum of Decision, "Development of Manned Space Station," undated; memorandum from Edwin Meese III and Robert McFarlane to the President, "Space Station Decision," December 10, 1983; memorandum from Fred Khedouri to Craig Fuller, "Space Station Decision Memorandum," December 12, 1983, all in Box 7, Outer Space Files, RRL.